SEC2_APM1110-FA1 GROUP_8-DABU, L; LIBRES, B

Github: https://github.com/dabulira20/Prob-ProbDi-FAs/tree/main/FA1

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Problem 1

```
Write the skewness program, and use it to calculate the skewness coefficient of the four
examination subjects in results.txt (results.csv). What can you say about these data?
Is Pearson's approximation reasonable?
```

```r
results <- read.csv("results.csv", na.strings = "NA")

# exact skewness (Equation 2.1)
exact_skewness <- function(x) {
  x <- x[!is.na(x)]
  n <- length(x)
  m <- mean(x)
  m2 <- sum((x - m)^2) / n
  m3 <- sum((x - m)^3) / n
  m3 / sqrt(m2)^3
}

# Pearson's approximate skewness: 3*(mean - median) / sd
pearson_skewness <- function(x) {
  x <- x[!is.na(x)]
  3 * (mean(x) - median(x)) / sd(x)
}

subjects <- c("arch1", "prog1", "arch2", "prog2")

skewness_results <- data.frame(
  arch1 = c(exact_skewness(results$arch1), pearson_skewness(results$arch1)),
  prog1 = c(exact_skewness(results$prog1), pearson_skewness(results$prog1)),
  arch2 = c(exact_skewness(results$arch2), pearson_skewness(results$arch2)),
  prog2 = c(exact_skewness(results$prog2), pearson_skewness(results$prog2))
)
rownames(skewness_results) <- c("Exact Skewness", "Pearson Skewness")
print(skewness_results)
```

```
##                       arch1      prog1     arch2      prog2
## Exact Skewness   -0.5070824 -0.3443763 0.4623552 -0.2807343
## Pearson Skewness -0.5896811 -0.6345275 0.5190494 -0.3854051
```

What can you say about these data? `arch1`, `prog1`, and `prog2` are negatively (left) skewed, meaning most students scored fairly well with a smaller tail of low scorers. `arch2` is positively (right) skewed instead, meaning most students clustered at lower scores with a few high scorers pulling the tail up. None of the subjects is heavily skewed — all four values are well under 1 in magnitude.

Is it a reasonable approximation? Pearson's formula gets the sign right for all four subjects, correctly flagging `arch2` as the only right-skewed one. The magnitudes are somewhat close but generally larger than the exact values (e.g. `prog1`: -0.63 vs -0.34), so it's a reasonable quick approximation but not a substitute for the exact formula when precision matters.

Problem 2

```
For the class of 50 students of computing detailed in Exercise 1.1, use R to
(a)  form the stem-and-leaf display for each gender, and discuss the advantages
of this representation compared to the traditional histogram;
(b)  construct a box-plot for each gender and discuss the findings.
```

Answers:

```r
# ------------------------------------------------------------------
# Purpose: Create stem-and-leaf displays comparing first-year Java
# programming exam scores between male and female students, and use
# them to visually compare score distributions by gender.
# ------------------------------------------------------------------

# Raw exam scores for the 23 female students.
female <- c(57, 59, 78, 79, 60, 65, 68, 71, 75, 48, 51, 55, 56, 41, 43,
            44, 75, 78, 80, 81, 83, 83, 85)
# Raw exam scores for the 27 male students.
male <- c(48, 49, 49, 30, 30, 31, 32, 35, 37, 41, 86, 42, 51, 53, 56, 42, 44,
          50, 51, 65, 67, 51, 56, 58, 64, 64, 75)

# Printed the labels, then generated the stem-and-leaf plot for each gender.
# stem() automatically splits each score into a "stem" (tens digit)
# and a "leaf" (ones digit) and prints the plot directly to console.
print("First-Year Java Programming Exam Scores of Female Students")
```

(a) Stem-and-Leaf Display by Gender

```
## [1] "First-Year Java Programming Exam Scores of Female Students"
```

```r
stem(na.omit(female))
```

```
##
##   The decimal point is 1 digit(s) to the right of the |
##
##   4 | 1348
##   5 | 15679
##   6 | 058
##   7 | 155889
##   8 | 01335
```

```r
print("First-Year Java Programming Exam Scores of Male Students")
```

```
## [1] "First-Year Java Programming Exam Scores of Male Students"
```

```r
stem(na.omit(male))
```

```
##
##   The decimal point is 1 digit(s) to the right of the |
##
##   3 | 001257
##   4 | 1224899
##   5 | 01113668
##   6 | 4457
##   7 | 5
##   8 | 6
```

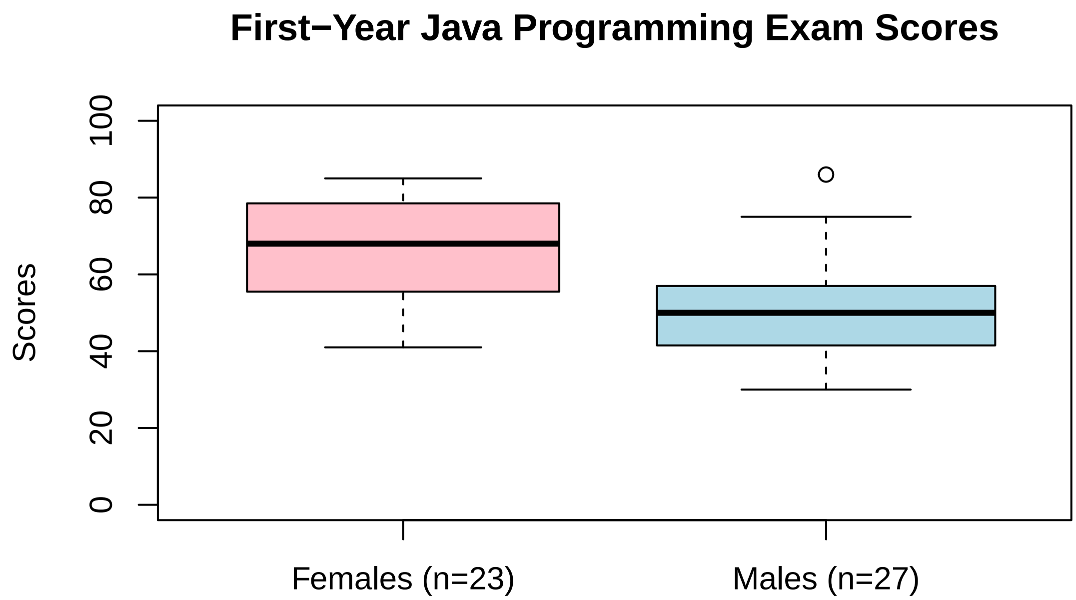
```r
# Note: na.omit() ensures the stem-and-leaf display is robust to any
# missing values in the data. Default stem() scale was reviewed and
# found appropriate for this data's range and sample size.
```

Advantages of Stem-and-Leaf over Histogram: A stem-and-leaf plot retains the actual data values while still showing the shape of the distribution. Unlike a histogram, you can read individual scores directly from the display, making it easier to spot outliers and check symmetry without losing information.

```r
# ---------------------------------------------------------------
# Purpose: Construct side-by-side box plots to visually compare the
# spread, median, and outliers of exam scores between female and
# male students.
# ---------------------------------------------------------------
# boxplot() draws one box per vector supplied (female, then male).
# names = labels each box; col = fills each box a different color
# for easier visual distinction between groups.

par(mar = c(5, 4, 4, 2) + 0.1)
boxplot(female, male,
        names = c("Females (n=23)", "Males (n=27)"),
        main = "First-Year Java Programming Exam Scores",
        ylab = "Scores",
        ylim = range(0, 100),
        col = c("pink", "lightblue"),
        varwidth = TRUE)
```

(b) Box Plots by Gender




```
# Note: varwidth = TRUE scales each box's width by its sample size,
# and axis labels display each group's n directly, so the unequal
# sample sizes (23 females vs. 27 males) are now visually accounted
# for. Outlier detection uses R's standard 1.5 x IQR (Tukey)
# method, which is a conventional and appropriate choice for
# this type of exploratory analysis.
```

Discussion of Findings:

- The box for females sits higher on the chart than the box for males, so females generally scored better on the exam.
- The female box looks taller than the male box, meaning female scores were more spread out, while male scores were more bunched or compressed together.
- There's a dot above the male box (an outlier), showing one male student scored much higher than the rest of the male group. The female box has no dots like this, meaning there were no extreme outliers in the female scores.
- Overall, the boxplot shows females scored higher on average but with more variation, while males scored lower but more consistently.

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